Acetylcysteine Prescribing and Administration Chart for 12-hr shortened N-acetylcysteine dosing schedule (SNAP protocol) – **RHC Glasgow**

Name: _____
Address: _____
DoB: _____
CHI: _____
Affix patient data label

Infusion 1 & 2 only

Please ensure that acetylcysteine is also prescribed on the patient's HEPMA Kardex.

Weight:……kgs
(NO NOT USE If patient <6 years of age)

Infusion 1	**Acetylcysteine 100mg/kg over 2 hours**									
Prescription						Preparation	Administration checks			
Date	Time	Dose (mL)	Diluent (200mL)	Infusion rate (mL/hr)	Prescriber's signature	Prepared/ Checked by	Date Time	Volume remaining (mL)	Volume infused (mL)	Checked by
Comments:			Stopped by:							
			Date:	Time	Signature					

Infusion 2	**Acetylcysteine 200mg/kg over 10 hours**									
Prescription						Preparation	Administration checks			
Date	Time	Dose (mL)	Diluent (1000mL)	Infusion rate (mL/hr)	Prescriber's signature	Prepared/ Checked by	Date Time	Volume remaining (mL)	Volume infused (mL)	Checked by
Comments:			Stopped by:							
			Date:	Time	Signature					

EXTENDED TREATMENT

- If extended treatment with acetylcysteine is required (see clinical guideline) then continue at the dose and infusion rate used for the infusion 2 and prescribe on the infusion 3 & 4 chart.
- Recheck bloods as per guideline 2hrs before the end of the infusions and refer to guideline regarding discontinuation of extended treatment.

Patients with an increase in INR and normal ALT

Both paracetamol and acetylcysteine treatment may cause an increase in INR in the absence of liver injury.

Patients who do not meet any of the criteria for continuation of acetylcysteine treatment but have an increase in INR of 0.4 or less (e.g. 1.1 to 1.5) AND have a normal ALT do not require further acetylcysteine.

Patients who have an increase in INR of 0.5 or more (e.g. 1.1 to 1.6) **without** an ALT rise - **STOP** acetylcysteine & recheck INR and ALT **after 4 - 6 hours**. If bloods show INR is unchanged or falling **AND** ALT is less than two times the upper limit of normal then no further treatment is required.

If the criteria above are not met - restart acetylcysteine at the dose and infusion rate used in the last treatment bag.

Acetylcysteine Antidote Adverse Effects – Features & Management

REACTION to acetylcysteine				COMPLICATIONS of paracetamol ingestion			
None ☐	Wheeze ☐			Abnormal liver function ☐		Encephalopathy ☐	
Flushing ☐	Hypotension ☐			Acute kidney injury ☐		Haemorrhage ☐	
Vomiting ☐	Other: ☐			Hypoglycaemia ☐		Other: ☐	
Rash ☐	Specify.....			Acidosis ☐		Specify.....	
Date and time of reaction		Initial		Date and time of reaction		Initial	

MANAGEMENT OF SIDE EFFECTS

- N-acetylcysteine may cause anaphylactoid reactions in 2% of cases with this protocol. Flushing, pruritus, rash, hypotension, angioedema, bronchospasm and vomiting are most common.
- Reactions can be managed by stopping the infusion. Consider chlorphenamine for flushing/itch, nebulised salbutamol if there is bronchospasm and ondansetron if there are GI side effects.
- **Restart the infusion once the reaction has resolved at half the rate to completion of infusion.**
- Previous reaction is **NOT** a contra-indication to N-acetylcysteine and cases should receive treatment if indicated. Reactions are now considerably less common with the 12-hour SNAP protocol compared to standard regimes.

Ondansetron oral or IV slow (over 2mins) injection (Nausea and vomiting) - Age 6 months-16 years

Body weight	Dose
Up to 10kg	2mg three times daily
10 - 40kg	4mg three times daily
41kg and above	8mg three times daily

Chlorphenamine ORAL (Rash and itch)

Age	Dose
1-23 months	1mg twice per day
2-5 years	1mg 4-6 hourly (maximum 6mg per day)
6-11 years	2mg 4-6 hourly (maximum 12mg per day)
12-16 years	4mg 4-6 hourly (maximum 24mg per day)

Chlorphenamine IV INJECTION (Rash and itch)

Age	Dose
1-5 months	250 micrograms/kg (maximum four times daily)
6 months - 5 years	2.5mg (maximum four times daily)
6 - 11 years	5mg (maximum four times daily)
12 - 16 years	10mg (maximum four times daily)